

1. Search for a specific word "apple" in a file and count it.

```
grep -o "apple" fruits.txt | wc -l
```

```
vboxuser@shivam:~/nov-2025$ ls -lh
total 720K
drwxrwxrwx 2 vboxuser vboxuser 4.0K Mar  3 10:33 bash-script
-rw-rw-r-- 1 nov-2025 nov-2025 661K Nov 22 19:18 file
-rw-rw-r-- 1 vboxuser vboxuser 2.6K Mar  2 22:15 file1
-rw-rw-r-- 1 vboxuser vboxuser 102 Nov 27 09:47 file1-backup-271125
dr--r--r-x 2 vboxuser vboxuser 4.0K Mar  2 20:36 file3.txt
-rw-rw-r-- 1 vboxuser vboxuser 19 Nov 19 10:10 file7
-rw-r----- 1 vboxuser vboxuser 599 Nov 23 04:30 fruits.txt
drwxrwxr-x 4 vboxuser vboxuser 4.0K Apr 29 16:06 github
-rw-r----- 1 vboxuser vboxuser 1.9K Nov 23 04:20 passwd-backup
drwxrwxr-x 3 vboxuser vboxuser 4.0K Apr 29 15:43 project
drwxrwxr-x 2 vboxuser vboxuser 4.0K Jan  1 22:02 Project2
-rw-rw-r-- 1 vboxuser vboxuser 36 Nov 22 05:38 softlink
-rw-r----- 1 vboxuser vboxuser  0 Nov 23 03:58 test-file-umask-after
-rw-rw-r-- 1 vboxuser vboxuser  0 Nov 23 03:58 test-file-umask-before
-rw-rw-r-- 1 vboxuser vboxuser  0 Nov 16 04:58 testing.txt
drwxr-x--- 2 vboxuser vboxuser 4.0K Nov 23 03:59 test-umask-after
drwxrwxr-x 2 vboxuser vboxuser 4.0K Nov 23 03:55 test-umask-before
-rw-r----- 1 vboxuser vboxuser 336 Mar  3 20:06 veg
vboxuser@shivam:~/nov-2025$ grep -o "apple" fruits.txt | wc -l
7
vboxuser@shivam:~/nov-2025$ |
```

2. Show current users logged in.

```
who
```

```
vboxuser@shivam:~/nov-2025$ who
vboxuser tty1          2026-04-30 13:23
vboxuser pts/0         2026-04-30 13:46 (10.0.2.2)
vboxuser@shivam:~/nov-2025$ |
```

3. Show total number of .txt files in current directory.

```
ls -l | wc -l
```

```
vboxuser@shivam:~/nov-2025$ ls -l | wc -l
18
vboxuser@shivam:~/nov-2025$ |
```

4. Show list of all system users from /etc/passwd.

cat /etc/passwd

```
vboxuser@shivam:~/nov-2025$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:./nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:./usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:./usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,./usr/lib/dhcpcd:/bin/false
messagebus:x:101:102:./nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:./usr/sbin/nologin
pollinate:x:102:1:./var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:./usr/sbin/nologin
syslog:x:103:104:./nonexistent:/usr/sbin/nologin
uidd:x:104:105:./run/uidd:/usr/sbin/nologin
tcpdump:x:105:107:./nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,./var/lib/tpm:/bin/false
landscape:x:107:109:./var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,./var/lib/usbmux:/usr/sbin/nologin
vboxuser:x:1000:1000:vboxuser:/home/vboxuser:/bin/bash
sshd:x:109:65534:./run/sshd:/usr/sbin/nologin
nov-2025:x:1001:1001:./home/nov-2025:/bin/sh
Sid-2025:x:1002:1002:./home/Sid-2025:/bin/sh
aug2025:x:1003:1003:Sidhu,07,1,2,0:/home/aug2025:/bin/bash
vboxuser@shivam:~/nov-2025$ |
```

5. Find how many times a word and each match should be on newline.

grep -o "word" filename.txt

echo "Total count: \$(grep -o "word" filename.txt | wc -l)"

```
vboxuser@shivam:~/nov-2025$ grep -o "apple" fruits.txt
apple
apple
apple
apple
apple
apple
apple
vboxuser@shivam:~/nov-2025$ echo "Total count: $(grep -o "apple" fruits.txt | wc -l)"
Total count: 7
vboxuser@shivam:~/nov-2025$ |
```

6. Show running processes and search for "root".

ps aux | grep root

```
vboxuser@shivam:~/nov-2025$ ps aux | grep root
root      1  0.8  0.2 22036 13200 ?        Ss   13:19   0:38 /sbin/init splash noprompt noshell automatic-ubiquity
root      2  0.0  0.0      0   0 ?        S    13:19   0:02 [kthreadd]
root      3  0.0  0.0      0   0 ?        S    13:19   0:00 [pool_workqueue_release]
root      4  0.0  0.0      0   0 ?        I<   13:19   0:00 [kworker/R-rcu_g]
root      5  0.0  0.0      0   0 ?        I<   13:19   0:00 [kworker/R-rcu_p]
root      6  0.0  0.0      0   0 ?        I<   13:19   0:00 [kworker/R-slub_]
root      7  0.0  0.0      0   0 ?        I<   13:19   0:00 [kworker/R-netns]
root     11  0.0  0.0      0   0 ?        I    13:19   0:00 [kworker/u10:0-ipv6_addrconf]
root     12  0.0  0.0      0   0 ?        I<   13:19   0:00 [kworker/R-mm_pe]
root     13  0.0  0.0      0   0 ?        I    13:19   0:00 [rcu_tasks_kthread]
root     14  0.0  0.0      0   0 ?        I    13:19   0:00 [rcu_tasks_rude_kthread]
root     15  0.0  0.0      0   0 ?        I    13:19   0:00 [rcu_tasks_trace_kthread]
root     16  0.0  0.0      0   0 ?        S    13:19   0:00 [ksoftirqd/0]
root     17  0.0  0.0      0   0 ?        I    13:19   0:02 [rcu_preempt]
root     18  0.0  0.0      0   0 ?        S    13:19   0:00 [migration/0]
root     19  0.0  0.0      0   0 ?        S    13:19   0:00 [idle_inject/0]
root     20  0.0  0.0      0   0 ?        S    13:19   0:00 [cpuhp/0]
root     21  0.0  0.0      0   0 ?        S    13:19   0:00 [cpuhp/1]
root     22  0.0  0.0      0   0 ?        S    13:19   0:00 [idle_inject/1]
root     23  0.0  0.0      0   0 ?        S    13:19   0:02 [migration/1]
root     24  0.0  0.0      0   0 ?        S    13:19   0:03 [ksoftirqd/1]
root     27  0.0  0.0      0   0 ?        S    13:19   0:00 [cpuhp/2]
root     28  0.0  0.0      0   0 ?        S    13:19   0:00 [idle_inject/2]
root     29  0.0  0.0      0   0 ?        S    13:19   0:02 [migration/2]
root     30  0.0  0.0      0   0 ?        S    13:19   0:00 [ksoftirqd/2]
root     33  0.0  0.0      0   0 ?        S    13:19   0:00 [cpuhp/3]
root     34  0.0  0.0      0   0 ?        S    13:19   0:00 [idle_inject/3]
root     35  0.0  0.0      0   0 ?        S    13:19   0:02 [migration/3]
root     36  0.0  0.0      0   0 ?        S    13:19   0:00 [ksoftirqd/3]
root     39  0.0  0.0      0   0 ?        S    13:19   0:00 [cpuhp/4]
root     40  0.0  0.0      0   0 ?        S    13:19   0:00 [idle_inject/4]
root     41  0.0  0.0      0   0 ?        S    13:19   0:00 [migration/4]
root     42  0.0  0.0      0   0 ?        S    13:19   0:03 [ksoftirqd/4]
root     48  0.0  0.0      0   0 ?        I    13:19   0:00 [kworker/u14:0-flush-8:0]
root     50  0.0  0.0      0   0 ?        S    13:19   0:00 [kdevtmpfs]
root     51  0.0  0.0      0   0 ?        I<   13:19   0:00 [kworker/R-inet_]
root     54  0.0  0.0      0   0 ?        S    13:19   0:00 [kauditd]
root     58  0.0  0.0      0   0 ?        S    13:19   0:00 [khungtaskd]
root     59  0.0  0.0      0   0 ?        S    13:19   0:00 [oom_reaper]
root     62  0.0  0.0      0   0 ?        I<   13:19   0:00 [kworker/R-write]
```

7. Show only .log files and sort them alphabetically.

ls *.log | sort

```
vboxuser@shivam:~/nov-2025$ ls *.log | sort
file1.log
file2.log
file3.log
file4.log
file5.log
vboxuser@shivam:~/nov-2025$ |
```

8. See disk space and filter "drivers".

df -h

```
vboxuser@shivam:~/nov-2025$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           479M  1.1M  478M   1% /run
/dev/sda2       25G   4.0G   20G  18% /
tmpfs           2.4G     0   2.4G   0% /dev/shm
tmpfs           5.0M     0   5.0M   0% /run/lock
tmpfs           479M   12K  479M   1% /run/user/1000
vboxuser@shivam:~/nov-2025$ |
```

9. Sort names alphabetically and remove duplicates.

sort -u fruits.txt

```
vboxuser@shivam:~/nov-2025$ sort -u fruits.txt

apple
banana
grape
kiwi
mango
melon
orange
peach
pear
plum
vboxuser@shivam:~/nov-2025$ |
```

10. Find all .log files modified in last 1 day.

find . -type f -name "*.log" -mtime -1

```
vboxuser@shivam:~/nov-2025$ find . -type f -name "*.log" -mtime -1
./file1.log
./file5.log
./project/Devops-Tools/Shell-Script/backup-and-cleanup-script/logs/script.log
./file2.log
./github/DevOps-Tool/backup-and-cleanup-script/logs/script.log
./file4.log
./file3.log
vboxuser@shivam:~/nov-2025$ |
```

11. Delete .tmp files older than 7 days.

find . -type f -name "*.tmp" -mtime +7 -delete

```
vboxuser@shivam:~/nov-2025$ find . -type f -name "*.tmp"
./file3.tmp
./file2.tmp
./file1.tmp
vboxuser@shivam:~/nov-2025$ find . -type f -name "*.tmp" -mtime +7 -delete
vboxuser@shivam:~/nov-2025$ |
```

12. Count how many .java files exist in a project.

find . -type f -name "*.java" | wc -l

```
vboxuser@shivam:~/nov-2025$ find . -type f -name "*.java" | wc -l
2
vboxuser@shivam:~/nov-2025$ |
```

